

Retail Data Pipeline (Walmart Data) – Full Documentation

Group : 5

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1. إدارة وتخطيط المشروع (Project Planning & Management)

Overview:

This project focuses on building a scalable **Retail Data Pipeline** that processes Walmart-like sales data from multiple sources (CSV, APIs, streaming) into a structured analytics system.

Objectives:

- Collect raw retail data (sales, customers, products)
- Clean and transform data (ETL/ELT process)
- Store in a data warehouse
- Enable analytics & dashboards

Scope:

- Batch processing (historical data)
 - Optional: real-time streaming (future work)
 - Data visualization (Power BI / Tableau)
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Project Plan (Timeline)

Duration: 6–8 weeks

Phase	Duration	Deliverables
Planning	Week 1	Proposal, requirements
Design	Week 2	ERD, architecture
Development	Week 3–5	ETL pipeline
Testing	Week 6	Test cases
Deployment	Week 7	Cloud/local deployment
Presentation	Week 8	Final demo

Task Assignment & Roles

Role	Responsibilities
Data Engineer	Build ETL pipeline
Data Analyst	Create dashboards
Backend Dev	API & integration
Project Manager	Timeline & coordination

Risk Assessment & Mitigation

Risk	Impact	Solution
Data quality issues	Wrong insights	Data validation layer
Pipeline failure	Data loss	Retry & logging system
Large data volume	Performance drop	Use distributed processing (Spark)
Time constraints	Incomplete project	Prioritize core features

KPIs (Key Performance Indicators)

- Pipeline execution time (< 5 min)
- Data accuracy (> 95%)
- System uptime (> 99%)
- Query response time (< 2 sec)
- Dashboard usage rate

2. مراجعة الأدبيات (Literature Review)

Feedback & Evaluation (Example)

- Strong pipeline structure and modular design
- Good use of ETL tools
- Clear documentation

Suggested Improvements

- Add real-time streaming (e.g., Apache Kafka)
- Improve data quality checks
- Add machine learning predictions
- Optimize query performance

Final Grading Criteria

Component	Weight
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Documentation	25%
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Implementation	35%
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Component	Weight
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Testing	20%
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Presentation	20%
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3. جمع المتطلبات (Requirements Gathering)

Stakeholder Analysis

Stakeholder	Needs
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Business Manager	Sales insights
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Data Analyst	Clean data
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Customers	Better services
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Developers	Stable system
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User Stories

- “As a data analyst, I want clean sales data to generate reports.”
- “As a manager, I want dashboards to track performance.”
- “As a system admin, I want logs to monitor failures.”

Functional Requirements

- Data ingestion (CSV, API)
- Data cleaning & transformation
- Data storage (warehouse)
- Dashboard visualization
- Logging & monitoring

Non-Functional Requirements

- Performance → fast processing
 - Security → data encryption
 - Scalability → handle big data
 - Reliability → fault tolerance
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4. تحليل وتصميم النظام (System Analysis & Design)

1. Problem Statement & Objectives

Problem:

Retail companies generate huge volumes of data that are difficult to process manually.

Solution:

Automated data pipeline to process and analyze data efficiently.

Objectives

- Automate data flow
- Ensure data accuracy
- Enable real-time insights (future)

Software Architecture

Architecture Style: Data Pipeline + Layered Architecture

Components:

- Data Sources
- ETL Layer
- Data Warehouse
- Visualization Layer

Tools Example:

- Python

- Apache Airflow
 - Apache Spark
 - Power BI
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2. Database Design & Data Modeling

ER Diagram (Entities)

- Customers
- Orders
- Products
- Stores

Example Tables

Orders Table

- OrderID (PK)
 - CustomerID (FK)
 - ProductID (FK)
 - Date
 - Amount
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3. Data Flow & System Behavior

Data Flow (DFD)

1. Input → Raw Data
2. Processing → Cleaning
3. Storage → Database

4. Output → Dashboard

Sequence Example

1. Upload data
2. ETL runs
3. Data stored
4. Dashboard updated

Activity Flow

- Start → Load data → Clean → Transform → Store → Visualize → End

State Example

- Raw → Processed → Stored → Analyzed

Class Example

Classes:

- DataLoader
- Transformer
- Validator
- DatabaseConnector

4. UI/UX Design

Wireframes

- Dashboard page
- Sales report page
- Filter panel

UI Guidelines

- Simple layout
- Clear charts

- Color coding (green = profit, red = loss)
 - Responsive design
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5. System Deployment & Integration

Technology Stack

Layer	Tech
Backend	Python
Processing	Spark
Orchestration	Airflow
Database	PostgreSQL
Visualization	Power BI

Deployment

- Local server or cloud (AWS / Azure)
- Scheduled jobs using Airflow

Component Diagram

- Data Source → ETL → DB → Dashboard
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6. Additional Deliverables

API Documentation (Optional)

Endpoints:

- /get-sales
- /get-customers

Testing & Validation

- Unit Testing → functions
- Integration Testing → pipeline
- UAT → dashboards

Deployment Strategy

- Use cloud services (AWS)
- Automate pipeline scheduling
- Scale using distributed systems